The National School of Artificial Intelligence
Computer Architecture
RISC-V Instruction Set

Lab 08: Files

Objectives: Files

After completing this lab, you will:

- Learn how to open, read from, and write into a file.

Files

The operating system manages files on the disk storage. It provides system calls to open, read from, and write to files. The RARS tool simulates some of the services of the operating system and provides the following system calls:

Service	a7	Arguments	Result
Open file	1024	a0 = address of null-terminated string containing the file name. a1 = 0 if read only a1 = 1 if write-only with create a1 = 9 if write-only with create and append	a0 = file descriptor a0 =-1 if error
Read from file	63	a0 = file descriptor a1 = address of input buffer a2 = maximum number of characters to read	a0 = number of characters read a0 = 0 if end-of-file a0=-1 if error
Write to file	64	a0 = file descriptor a1 = address of output buffer a2 = number of characters to write	a0 = number of characters written a0 =-1 if error
Close file	57	a0 = file descriptor	

Table 1: Files system calls

Here is a RISC-V program that writes a string to a file named MyFile.txt:

```
.data
fout: .asciz "My_File.txt" # filename for output
buffer: .asciz "The quick brown fox jumps over the lazy dog."
.text
  # Open (for writing) a file that does not exist
  li    a7, 1024      # system call for open file
  la    a0, fout      # output file name
```

```
li    a1, 1         # Open for writing (flags are 0: read, 1: write)
ecall               # open a file (file descriptor returned in a0)
mv    s6, a0        # save the file descriptor
# Write to file just opened
li    a7, 64        # system call for write to file
mv    a0, s6        # file descriptor
la    a1, buffer    # address of buffer from which to write
li    a2, 70        # hardcoded buffer length
ecall               # write to file
# Close the file
li    a7, 57        # system call for close file
mv    a0, s6        # file descriptor to close
ecall               # close file
```

Listing 1: Files

Lab Assignment

1. Given the file dna.txt, count the occurrences of characters C and G in the text file, and the length of the original string. Use a message box to display the result.
2. Write a RISC-V program to copy an input text file into an output file. The input and output file names should be entered by the user. If the input file cannot be opened, print an error message.